

Answer Key & Teacher Notes

Session 1 – Range and IQR • Session 2 – Comparing Variability

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Range and IQR

TALK PROMPTS

- Which number describes only the middle half of the data, and why?
- Why does one very large value change the range but not the IQR?

PARTNER TASK

Camp Hikes. One partner finds the five-number summary of the seven hikes; the other computes the range and the IQR from it. Swap roles with Bake Sale and compare which measure each partner trusts more.

WATCH FOR

- Computing the range (max – min) when the IQR is asked.
- Subtracting Q1 from the max, or the min from Q3, instead of Q3 – Q1.
- Thinking a new extreme value changes the IQR.

SESSION 2 • TEACHER NOTES

Comparing Variability

TALK PROMPTS

- Which measure convinced you that one class was more consistent, and why?
- When could two groups have the same range but very different IQRs?

PARTNER TASK

Phone Batteries. One partner finds the range and IQR for Phone 1; the other does Phone 2. Compare and agree on which phone is steadier, using both measures.

WATCH FOR

- Comparing the range of one group with the IQR of the other.
- Thinking a greater IQR always means a greater range.
- Calling the group with more values the more spread out group.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.SP.B.5c	1	1	1	3	2	2	3	3	3	1	1
S1 • Form B	6.SP.B.5c	1	1	1	3	2	2	3	3	3	1	1
S2 • Form A	6.SP.B.5c	2	1	3	3	3	3	2	3	3	1	1
S2 • Form B	6.SP.B.5c	2	1	3	3	3	3	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

range: The greatest value **minus** the least value.

interquartile range (IQR): Q3 minus Q1; the spread of the **middle half**.

extreme value: A value far from the others; it changes the **range** a lot.

- Range and IQR both measure **spread**, or variability.
- Changing only the greatest value changes the range but not the **IQR**.

Example: Order: 5, 8, 10, 12, 15, 18, 20. Range = $20 - 5 = 15$. Q1 = **8** and Q3 = **18**. Answer: IQR = $18 - 8 = 10$ km

YOU TRY

- Order: 4, 6, 7, 9, 11, 13, 15. a. 11 b. $13 - 6 = 7$
- $12 + 25 = 37$

EXIT TICKET

- Order: 3, 5, 7, 8, 10, 12, 14. a. 11 b. $12 - 5 = 7$
- $30 - 6 = 24$; $22 - 10 = 12$
- B** (Only the range changes)

PRACTICE FORM A

- Order: 5, 8, 10, 12, 15, 18, 20. a. $20 - 5 = 15$ b. 8 c. 18 d. $18 - 8 = 10$
- Order: 48, 51, 55, 59, 62, 64, 70. a. $70 - 48 = 22$ b. $64 - 51 = 13$
- Order: 55, 61, 66, 70, 72, 79, 88. a. $88 - 55 = 33$ b. $79 - 61 = 18$
- a. 21; 83 b. 15; 15 c. No. Only the range grows; Q1 and Q3 do not change, so the IQR stays 15.
- a. $7 + 38 = 45$ b. $32 - 15 = 17$
- a. $44 - 20 = 24$ b. $40 - 24 = 16$ c. $44 - 20 = 24$ d. $42 - 22 = 20$. Same range, but D's middle half is more spread out.
- a. $62 - 11 = 51$ b. $19 - 14 = 5$ c. the IQR (the \$62 wage stretches the range)
- a. A: range 40, IQR 14; B: range 20, IQR 7 b. Bar B: its IQR is smaller. c. One unusual day changes the min or max, so it changes the range a lot but not the IQR.
- Tobias found the range ($36 - 6 = 30$). The IQR is Q3 - Q1 = $30 - 10 = 20$.
- B** (greatest - least)
- B** (14)

Self-check word EXTREME

PRACTICE FORM B

- Order: 6, 9, 11, 14, 17, 19, 22. a. $22 - 6 = 16$ b. 9 c. 19 d. $19 - 9 = 10$
- Order: 52, 55, 58, 62, 66, 69, 74. a. $74 - 52 = 22$ b. $69 - 55 = 14$
- Order: 50, 58, 62, 66, 68, 75, 84. a. $84 - 50 = 34$ b. $75 - 58 = 17$
- a. 24; 80 b. 13; 13 c. No. Only the range grows; Q1 and Q3 do not change, so the IQR stays 13.
- a. $9 + 41 = 50$ b. $37 - 18 = 19$
- a. $40 - 10 = 30$ b. $35 - 15 = 20$ c. $40 - 10 = 30$ d. $39 - 11 = 28$. Same range, but D's middle half is more spread out.
- a. $70 - 12 = 58$ b. $21 - 15 = 6$ c. the IQR (the \$70 wage stretches the range)
- a. A: range 30, IQR 11; B: range 46, IQR 22 b. Bar A: its IQR is smaller. c. One unusual day changes the min or max, so it changes the range a lot but not the IQR.
- Marisol found the range ($40 - 5 = 35$). The IQR is Q3 - Q1 = $31 - 9 = 22$.
- B** (middle half)
- A** (17)

Self-check word MEASURE

GUIDED NOTES

consistent: Values are close together; the spread is **small**.

measure of variability: A number that tells how **spread out** data are.

extreme value: A value far from the rest; it stretches the **range**.

- Compare the **same** measure for both groups. The smaller value means more consistent.
- When one value is far from the rest, trust the **IQR** more than the range.

Example: Ranges: $18 < 35$, so R is **less** spread overall. IQRs: $12 < 25$, so R's **middle half** is closer together. Answer: Class **R** is more consistent.

YOU TRY

1. a. 12 b. 6; Set Q
2. Set A (its range is much larger than its IQR)

EXIT TICKET

1. a. 12 b. 4 (Y is more consistent)
2. Team 1 (IQR $6 < 14$)
3. **B** (IQR)

PRACTICE FORM A

1. Class R: range $88 - 70 = 18$, Q1 73, Q3 85, IQR 12. Class S: range $95 - 60 = 35$, Q1 65, Q3 90, IQR 25. More consistent: Class R
2. a. $50 - 30 = 20$ b. $46 - 33 = 13$ c. $45 - 34 = 11$ d. $43 - 37 = 6$. Machine 2 is steadier.
3. Not right. V's IQR is 20, W's IQR is 5: V's middle half is much more spread out.
4. Not right. The middle halves match, but T's range is 24 and U's range is 88, so U is much more spread out overall.
5. The IQR. The one extreme value stretches the range, but it does not affect Q1 or Q3, so the IQR describes the typical spread.
6. a. Set N b. Set M (its range is much larger than its IQR)
7. a. $26 - 14 = 12$ b. $24 - 16 = 8$ c. $24 - 17 = 7$ d. $23 - 18 = 5$. Phone 2 is more consistent.
8. a. $13 - 9 = 4$ b. $18 - 3 = 15$ c. Bus 1 (smaller IQR) d. Bus 2 is sometimes on time, but its late times vary a lot (up to 40 min). Bus 1 is late by about the same amount every day, so you can plan for it.
9. One measure does not decide the other. Example: Set A = 2, 4, 6, 10, 14, 16, 18 has IQR 12 and range 16, while Set B = 0, 8, 9, 10, 11, 12, 40 has IQR 4 and range 40. Check both measures.
10. **B** (closer together)
11. **A** (an extreme value)

Self-check word VARYING

PRACTICE FORM B

1. Team J: range $83 - 64 = 19$, Q1 67, Q3 80, IQR 13. Team K: range $92 - 55 = 37$, Q1 60, Q3 88, IQR 28. More consistent: Team J
2. a. $52 - 28 = 24$ b. $48 - 31 = 17$ c. $49 - 38 = 11$ d. $47 - 40 = 7$. Machine 2 is steadier.
3. Not right. V's IQR is 25, W's IQR is 6: V's middle half is much more spread out.
4. Not right. The middle halves match, but T's range is 22 and U's range is 83, so U is much more spread out overall.
5. The IQR. The one extreme value stretches the range, but it does not affect Q1 or Q3, so the IQR describes the typical spread.
6. a. Set M b. Set N (its range is much larger than its IQR)
7. a. $21 - 8 = 13$ b. $19 - 10 = 9$ c. $18 - 12 = 6$ d. $17 - 13 = 4$. Tablet 2 is more consistent.
8. a. $20 - 5 = 15$ b. $13 - 10 = 3$ c. Bus 2 (smaller IQR) d. Bus 1 is sometimes only 2 min late, but its times vary a lot (up to 35 min). Bus 2 is late by about the same amount every day, so you can plan for it.
9. One measure does not decide the other. Example: Set A = 2, 4, 6, 10, 14, 16, 18 has IQR 12 and range 16, while Set B = 0, 8, 9, 10, 11, 12, 40 has IQR 4 and range 40. Check both measures.
10. **B** (Set Y)
11. **B** (spread of the middle half)

Self-check word COMPARE

SHEET 1 • RANGE AND IQR

1. a. $21 - 6 = 15$ b. $60 - 38 = 22$ c. $31 - 18 = 13$ d. $56 - 40 = 16$ e. $90 - 68 = 22$ f. $25 - 7 = 18$
2. a. $30 - 6 = 24$ pages b. $22 - 10 = 12$ pages
3. Ordered: 28, 35, 41, 47, 52, 55, 60. a. $60 - 28 = 32$
b. Median 47; $Q1 = 35$, $Q3 = 55$; $IQR = 55 - 35 = 20$
4. a. $14 + 29 = 43$ b. $20 + 11 = 31$
5. Half of the range is $32 \div 2 = 16$, but the IQR is 20. The IQR depends on $Q1$ and $Q3$, not on the min and max.

SHEET 2 • COMPARING VARIABILITY

1. a. 12 b. 6 (Set Q is more consistent)
2. Player 1: range $16 - 8 = 8$, IQR $14 - 10 = 4$. Player 2: range $22 - 2 = 20$, IQR $18 - 6 = 12$. Player 1 is more consistent (both medians are 12).
3. Ranges: A: $80 - 70 = 10$; B: $95 - 55 = 40$. More consistent: Group A (smaller range; both medians are 75)
4. a. Route 1: $40 - 5 = 35$; Route 2: $25 - 8 = 17$ b. Route 1: $18 - 12 = 6$; Route 2: $20 - 10 = 10$
5. Kwame. Route 2's range is smaller (17 vs. 35), but Route 1's range is large only because of one extreme ride (40 min). Route 1's middle half is closer together (IQR 6 vs. 10), and the IQR is the better measure when there is an extreme value.
6. **A** (Class A)
7. **A** (6)